

Technology Stack

Date	3 july 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	2 Marks

Define Your Technology Stack:

The technology stack for the HDI Prediction System consists of modern web technologies, machine learning frameworks, and database tools. It includes a responsive frontend, a Python-based backend, a MySQL database, and Scikit-learn for prediction. These technologies were selected to ensure accurate predictions, secure data management, easy maintenance, and a user-friendly experience.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	Generate predictions quickly.	Response time < 3 seconds

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Support multiple users simultaneously.	Supports 500+ users
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	Protect user data using secure storage.	Encrypted data and secure login

4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	System should be available whenever needed.	99% uptime
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Simple and user-friendly interface.	Easy navigation and responsive UI
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Easy maintenance and future model updates.	Modular and maintainable code